

Impact of Gas Line Absorption in Long-Haul transmission of 134 GBd Signals in C-Band over Low Loss and Low IMI HCF

Rajiv Boddeda^(1*), Carina Castineiras Carrero⁽¹⁾, Haik Mardoyan⁽¹⁾, Amirhossein Ghazisaeidi⁽¹⁾, Ruby S. B. Ospina^(1,2), Peng Li⁽³⁾, Shuhai Li⁽³⁾, Lei Zhang⁽³⁾, Jie Luo⁽³⁾, and Jérémie Renaudier⁽¹⁾

⁽¹⁾ Nokia Bell Labs, Optical Transmission Dept., Massy, France; *rajiv.boddeda@nokia-bell-labs.com

⁽²⁾ Nokia Bell Labs, New Providence, New Jersey, USA

⁽³⁾ State Key Laboratory of Optical Fiber and Cable Manufacture Technology, Yangtze Optical Fibre and Cable (YOFC), Wuhan, China.

Abstract We report on the C-band WDM transmission of 134-GBd channels using DP-16QAM and DP-64QAM over long-haul distances. Using a recirculating loop comprising of a single span made of 266-km of GTA-ST-HCF, we show the impact of gas line absorption on SC and MC modulation schemes. ©2026 The Author(s)

Introduction

The unprecedented spread of AI applications, along with the rise of IP services and cloud computing poses significant challenges to the current optical networks architecture currently supported by silica-based standard single-mode fibres (SSMFs), digital coherent technologies, ultra-wideband systems, and pluggable optics. In this context, disruptive technologies are being explored to increase the capacity of optical systems [1,2]. Anti-resonant hollow-core fibres (AR-HCFs), designed to confine light in an air-filled hollow core, are being considered as a strong alternative for next generation optical networks by achieving ultra-low loss while overcoming the nonlinearity and latency limitations of SSMFs at the same time [3,4]. Leveraging the weak nonlinear stimulated Raman scattering (SRS), several multi-band demonstrations over HCFs have been recently carried out, reporting different transmission distances and total capacity over C+L [5] and S+C+L [6]. Currently, long-distance optical transmission over HCFs is limited by both gas line absorption (GLA) and the non-negligible presence of higher order modes (HOM) during the propagation. The latter effect causes inter-modal interference (IMI) and strongly degrades the signal quality [7,8]. IMI induces a notable frequency-selective fading that limits the system

performance and the reach. Leveraging transmitter and receiver digital signal processing (DSP) has been proposed to enhance transmission tolerance against IMI, thus improving the channel capacity and the reach [8]. Meanwhile the reduction of IMI in HCFs can be handled through fiber design and fabrication strategies aiming at suppressing HOM and minimizing bending sensitivity. Recently a newly designed AR-HCF has been proposed to lower IMI while maintaining low loss characteristics, enabling a record wavelength division multiplexed (WDM) transatlantic transmission [9]. In that work, while IMI is shown to be significantly reduced, the presence of GLA peaks in the C band appears as the main limitation for long haul transmission systems.

In this paper, we experimentally compare the performance of single-carrier (SC) and digital multi-carrier (MC) multiplexing over long-haul distances through a low-loss low IMI AR-HCF link. Based on a recirculating loop with a 266-km span of gap tube assisted support tube-hollow core fiber (GTA-ST-HCF), we show the induced penalty incurred due to GLA peaks beyond 1000 km for both modulation schemes, covering a vast majority of long-haul terrestrial links [10].

Experimental setup

Fig. 1a depicts the experimental set-up used for

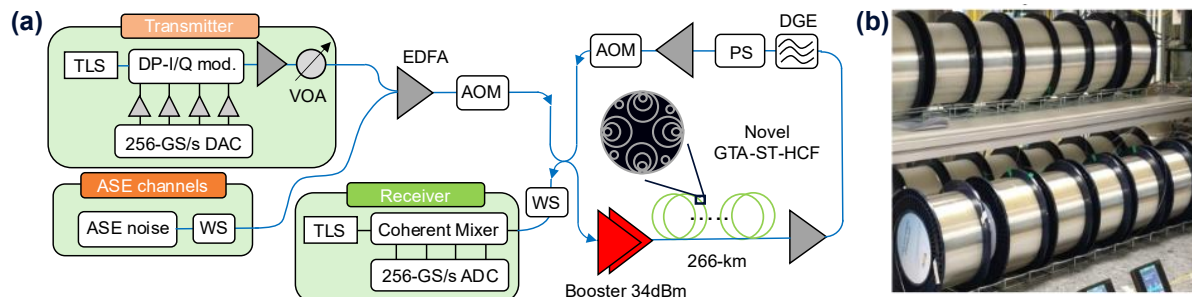


Fig. 1. (a) Experimental recirculating loop testbed composed of a single span made of a high-power booster amplifier followed by 266-km of GTA-ST-HCF. TLS: Tunable Laser Source; WS: Wave Shaper; PS: polarization scrambler; AOM: acousto-optic modulator. VOA: Variable Optical Attenuator. EDFA: Erbium doped fiber amplifier. DGE: Dynamic Gain Equalizer. (b) 23 spools of varying length between 10-15 kms were spliced together to make 266-km span.

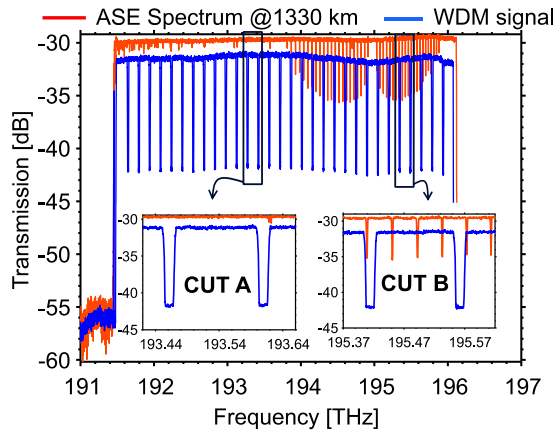


Fig. 2: The ASE transmission spectrum after 1330km is shown in red. In blue, the WDM signal spectrum is shown. In the inset, we show the spectrum of two channels at 193.538 THz and 195.488THz without and with GLA.

reproducing ultra-long-haul transmissions. The transmitted signal is made of a fully loaded WDM comb composed of carved amplified spontaneous emission (ASE) noise over 4.8 THz plus one channel under test (CUT). The CUT is made of a C-band tunable laser source (TLS) modulated with dual polarization 16-QAM signal using either a single carrier or multiple subcarriers with a total rate of 134 GBaud. The digital waveforms are generated by digital-to-analog converters (DACs) from a commercially available arbitrary waveform generator operating at 256-GS/s with a 10-dB bandwidth of 70-GHz. The digital signal is Nyquist shaped (root-raised-cosine (RRC), roll-off 0.01 and 0.06) for all symbol rates. The generated WDM comb signal is sent into the recirculating loop and is first amplified using a carefully chosen 34 dBm amplifier with up to 35 dB gain and a maximum noise figure of 4.68 dB in the C-band. The amplified signal is launched at 34 dBm to the ultra-long span of 266 km of GTA-ST-HCF. In Fig. 1b, we show the 266 km span composed of 23 spools of varying length between 10-15 km spliced together and terminated with SMF-HCF adapters on both ends. As shown in the inset of Fig. 1, the GTA-ST-HCF combines a support tube architecture that enables ultra-low confinement loss with gap tube assisted modal filtering mechanism, thereby achieving both low attenuation and high modal purity. The estimated IMI of this

fiber is -68.8 dB/km [9].

At the end of the loop, the WDM comb signal is amplified and passed through a dynamic gain equalizer followed by a polarization scrambler. At the receiver side, after a given number of round trips inside the loop, the CUT is extracted from the WDM comb by a wave shaper configured as a bandpass filter. The CUT is then received by a standard coherent receiver front-end and sampled at 256-GS/s using a 110-GHz bandwidth real-time oscilloscope. For each measurement, a set of 5 waveforms is stored, each recorded waveform consisting of 16 million samples which are processed offline. The standard DSP suite consists of matched filter, chromatic dispersion compensation, complex 2x2 MIMO adaptive equalization, frequency offset and phase compensation, and least-mean square post-equalizer to mitigate for transmitter I/Q imbalances. We finally compute the signal-to-noise ratio (SNR), the normalized generalized mutual information (NGMI), and the achievable information rate (AIR).

In Fig. 2, we show the ASE signal after ~ 1330 km measured using an optical spectrum analyser and the signal spectrum with 150 GHz spacing. In the inset, we show the two wavelengths considered for this study, CUT A at 193.538 THz and CUT B at 195.488 THz. The former one was chosen in the absence of GLA peaks to study the impact of IMI, and the latter one was chosen in the GLA region with the maximum number of peaks possible in the signal i.e., ~ 3 . To avoid GLA spectral notches while keeping spectral efficiency high, one can adapt the channel baud rate (and therefore occupied bandwidth) per given spectral grid [9,11] but in this study, we fixed the channel grid to analyse the GLA impact on large baud rate systems.

Multi Carrier Transmission

We evaluated the performance of digital MC with up to 16 subcarriers for long-haul transmission. First, we studied the impact of IMI on one channel located at 193.538 THz i.e., CUT A, outside the GLA peaks regions. We limited our study to DP-16-QAM at 134 Gbaud using a RRC roll-off of

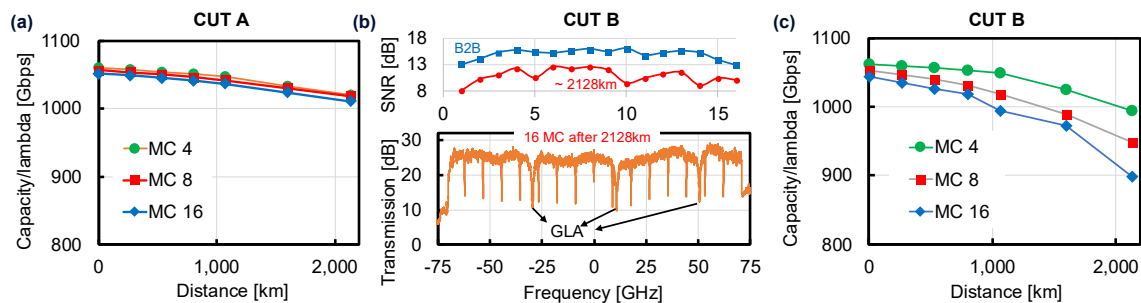


Fig. 3: a) Performance of MC as a function of distance without GLA b) (top) SNR for each subcarrier in B2B and at ~ 2128 km (bottom) The received spectrum of 16 MC (c) The capacity for different MC as a function of distance in the presence of GLA.

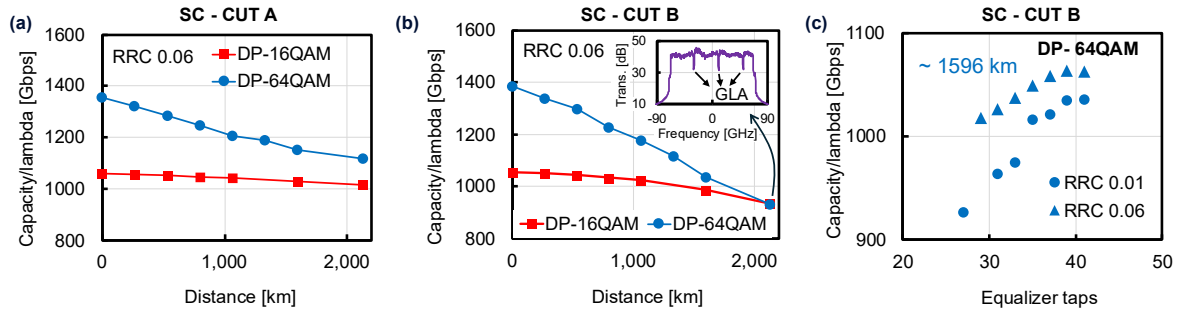


Fig. 4: Single carrier transmission a) Capacity per lambda vs distance for DP-16QAM and DP-64QAM b) In the presence of GLA, Capacity per lambda is shown as a function of distance. In the inset, we show the optical spectrum with GLA peaks after ~2128 km. c) Impact of the equalizer taps is estimated in the presence of GLA for different roll-offs at 1596 km.

0.06. In Fig. 3a, the AIR as a function of the distance for MC 4, 8 and 16 subcarriers are presented. We estimate an AIR reaching as high as 1000 Gbps per lambda up to 2000 km enabled by the low loss of the GTA-ST-HCF and low noise figure of the HPOA in the C-band. Moreover, we don't observe any increase in the performance with the number of subcarriers as the IMI of the fiber is relatively low compared to what has been shown in [8].

Then to study the impact of GLA, we consider the CUT B at 195.488 THz. We capture the optical spectrum of the 16 MC signal after 8 round trips i.e., 2128 km, where we observe the impact of GLA reaching as high as 10 dB as shown in Fig 3b (bottom). In Fig. 3b (top), the SNR for each of the subcarriers is presented. From Fig. 3b, the impact of the GLA over three subcarriers marked with the arrows is clearly observed. The SNR penalty for these subcarriers remains below 2 dB compared to their neighboring subcarriers, even though they experience significant spectral distortion. This minimal penalty is attributed to the equalizer in the DSP, which effectively compensates for the phase distortions caused by GLA peaks [12]. In Fig. 3c, we show the achievable capacity per lambda for various number of subcarriers. Contrary to the absence of GLA, here we observe that the capacity drops below 1000 Gbps after 1000 km. This can be attributed primarily to more localized distortions and stronger penalty at subcarriers aligned with absorption peaks that the receiver equalizer cannot compensate for. A more effective approach would involve pre-emphasizing the signal at the transmitter. However, in the absence of such measures, reducing the number of subcarriers is a better alternative, as it facilitates a more efficient allocation of symbol energy across a broader bandwidth.

Single Carrier Transmission

Now, we assessed the performance of single carrier at 134 Gbaud with a roll-off of 0.01 and 0.06. Unlike MC transmission, here we also consider DP-64QAM to further increase the maximum AIR. First, the RRC roll-off is set to 0.06. In Fig 4a, we

show the performance of the CUT A at 193.538 THz for both modulation formats. We observe that with DP-64QAM, we can transmit up to 1.2 Tbps up to 1000 km. With DP-16QAM, the transmission is limited by the maximum achievable information rate, and a notable reduction only starts to be observed at distances larger than 2000 km. In Fig. 4b, we show the impact of GLA by plotting the capacity per lambda as a function of distance. In the inset of Fig. 4b, we show the received signal spectrum showing the three GLA lines. We observe a more rapid deterioration with DP-64QAM compared to DP-16QAM reaching the same capacity at ~2000 km. Nevertheless, 1.2 Tbps could be transmitted up to 800 km using DP-64QAM, 1Tbps up to 1500 km and 800 Gbps per lambda even after 2000 km in the presence of GLA. In Fig. 4c, we show the impact of the equalizer length for different values of roll-off, 0.01 and 0.06 on CUT B. We observe that with a larger roll-off, fewer than 30 taps are sufficient to overcome the impact of GLA peaks and transmit 1Tbps at 1596 km. As expected, as we reduce the roll-off to 0.01, we would need more taps for the same capacity. However, from Fig. 4c, an equalizer with less than 40 taps is still sufficient to transmit 1Tbps at 1500 km. These results indicate that denser spectral packing is feasible, enabling a reduction in channel spacing from 150 GHz to 137.5 GHz while maintaining 1 Tbps transmission even in the presence of GLA.

Conclusions

We experimentally investigate long-haul transmission through a low-loss low IMI GTA-ST-HCF link using a recirculating loop comprising a single 266-km span of GTA-ST-HCF. We observe that a reduced number of subcarriers demonstrates greater resilience against GLA peaks. Our findings indicate that 800 G transmission is achievable over distances exceeding 2000 km using both SC and MC schemes. Furthermore, we demonstrate that capacities of up to 1.2 Tbps are feasible over 800 km, while 1 Tbps can be achieved over 1500 km. Notably, the equalizer complexity remains manageable even in the presence of GLA peaks and small roll-off factors.

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